

Applied A.I. Solutions

Foundations of Data Management

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DOCUMENT AND CONTENT MANAGEMENT

The DAMA Wheel



INTRODUCTION

- Document and Content Management entails **controlling** the capture, storage, access, and use of data and information stored **outside** relational databases
- Its focus is on maintaining the quality, security, integrity of and enabling **access to documents and other unstructured information**

DCM Framework

Definition

Planning, implementation, and control activities for lifecycle management of data and information found in **any form or medium**

Principles:

1. Everyone must create, use, retrieve, and dispose of records **in accordance** with the established policies and procedures
2. **Experts** in handling of Records and Content **should be fully engaged in policy and planning, and should comply with regulations and best practices**

ARMA International <https://www.arma.org/>

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- A set of **General Acceptable Recordkeeping Principles** (ARMA International / GARP, 2009) describes how business Records should be maintained, and provides a Governance Framework with associate metrics. ARMA International <https://www.arma.org/>

ARMA's Principles

- Accountability
- Integrity
- Protection
- Compliance
- Availability
- Retention
- Disposition
- Transparency

Drivers

- Regulatory compliance
- Ability to respond to litigation and
Ability to quickly respond to e-discovery requests (Big Data)
- Business continuity requirements
- Improve document management efficiencies

Goals

1. Comply with **legal obligations**, customer expectations on Records Management
2. Ensure effective storage, retrieval, use of documents and content
3. Ensure integration capabilities between structured, unstructured content



Business Drivers

Inputs

- Business strategy
- IT strategy
- Legal retention requirements:
 - Text file
 - Electronic format file
 - Printed paper file
 - Social media stream



Activities

1. Plan for Lifecycle Management
2. Create **Content Handling Policies**, including e-discovery
3. Define **Information Architecture**
4. Manage the Lifecycle
5. Publish Content



Deliverables

- Content and Records Management Strategy
- Policy and Procedures
- Content Repository
- Managed Records in many media formats
- Audit Trails and Logs

Essential Concepts

- **Content**
 - Content Management
 - Content Metadata
 - ✓ Format
 - ✓ Search-ability
 - ✓ Self-documentation
 - ✓ Existing Patterns
 - ✓ Content Subjects
 - ✓ Requirements

- **Controlled Vocabularies**

A Controlled Vocabulary is a **defined list of allowed terms** used to index, categorize, tag, sort, and retrieve content through browsing and searching

Constitute a type of **Reference Data or Metadata**.

Special cases:

- Vocabulary Views and Micro-controlled Vocabulary
- Term Management - Term and Pick Lists

ANSI/NISO Z39.19-2005, American standard, guidelines for the Construction, Format and Management of Monolingual Controlled Vocabularies

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- **Types of Controlled Vocabularies – cont'd**

Thesauri:

1. A **thesaurus** is used to map ambiguous words (terms) to discrete meanings (concepts).
2. It helps resolve synonyms and related terms by grouping them under a single heading.
3. For example, terms like “happy,” “sad,” and “excited” could all fall under the concept of “moods” or “emotions.”
4. Thesauri are part of **controlled vocabularies** and aid in consistent terminology usage.

- **Types of Controlled Vocabularies – cont'd**

Taxonomies:

1. **Taxonomies** organize content into hierarchical structures for filing and browsing.
2. They resemble a tree data structures, where top-level categories have sub-categories.
3. Used for grouping things, such as organizing digital libraries.
4. Each level deeper in the hierarchy becomes more specific (narrower).

- **Types of Controlled Vocabularies – cont'd**

Ontologies:

1. **Ontologies** describe interrelated concepts within a specific domain.
2. They aim to thoroughly represent a bounded world (domain) with a schema.
3. For instance, an ontology for car sales might define relationships between Cars, Companies, Make/Models, and Years.
4. Ontologies are used for **knowledge representation** and provide a blueprint for organizing data.

- **Documents** communicate or share information and knowledge. Documents are electronic or paper objects that contain instructions for tasks, requirements and logs
- **Records** serve as evidence of business activities, decision-making, and regulatory compliance. Only a subset of documents will be designated as Records

References: ISO 9000/9001, ISO 15489, Canada's Bill 198

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- **Document Management**

- Inventory
- Policy
 - ✓ Classification
 - ✓ Storage
 - ✓ Retrieval and Circulation
 - ✓ Preservation and Disposal

- **Record Management**

- Content
- Context (Metadata)
- Permanency
- Timeliness
- Structure

References: ISO 9000/9001, ISO 15489, Canada's Bill 198

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- **Litigation Response Playbook**
 - defines the target environment for **e-discovery**
 - assess if gaps exist between current and target environments
- **Litigation Response Data Map**
 - defines a comprehensive **catalog** of information systems and their uses, the information contained, retention policies, and other DLC characteristics.

- **Metrics**

- a) **Records Management**

- % of total documents per user identified as corporate records
 - % of total stored records with proper retention rules applied

- b) **E-discovery**

- Cost reduction
 - Efficiency gained in collecting information ahead of time

